Session 3 · Thursday 3 September 2026 · Every line to write, in order. Thirty-seven minutes of board across eight segments.

Board notes

Print this and carry it. The deck records what is *projected*; this records what gets *written*, which is the half of this session with the highest load on the room and no other artefact.

Thirty-seven minutes across eight segments, all before minute 74. Segments 1 and 5 each gave up a minute so ConcepTest 1 could have the four it needs for vote–argue–vote; the ConcepTest then moved to 33–37, hard against the pause.

Conventions. ASK — put it to the room before writing the next line, and the answer to expect. **CHECK** — do it out loud; doing them is the habit being taught. **IF ASKED** — the question that always comes. **CUT** — drop this first if you are behind, decided now rather than at minute 70.

Running total of board time is in the margin of each heading. If you are more than three minutes behind at the 37-minute mark, take the first CUT.

---

1 · 8–13 min — Where mass action comes from

Board 1 of 8 · 5 min · cumulative 5

Draw a box. Put two dots labelled A and three labelled B in it.

> ***ASK:*** *How many distinct A–B pairs can meet? Expect a beat, then "six." Write* ***2 × 3 = 6****.*

Add one more A.

> ***ASK:*** *Now? — "nine." Write* ***3 × 3 = 9****.*

Now write:

$$\text{encounters per second} \;\propto\; [\mathrm{A}]\,[\mathrm{B}]$$

Say: the proportionality is *counting pairs*. Nothing chemical has happened yet.

Then the rate law:

$$v \;=\; k\,[\mathrm{A}][\mathrm{B}]$$

> *Say explicitly what k absorbs: how often an encounter has enough energy, the right orientation, a catalyst present, the solvent, the temperature.* ***Every assumption in this model is hiding in that one letter.***

Then the coefficient point. Write the dimer:

$$\mathrm{P} + \mathrm{P} \to \mathrm{P}_2 \qquad v = k[\mathrm{P}]^2$$

> *Say: the stoichiometric coefficient enters the* ***flux as a power*** *and the* ***balance as a multiplier****. Same number, two different jobs. Confusing them is exercise E4 and it is the single most common algebra error in the course.*
>
> ***IF ASKED*** *"why not $2[\mathrm{P}]$?" — because you are counting pairs of P, and the number of pairs among n objects goes as n^2, not $2n$. Go back to the dots.*

CUT: the dimer paragraph — repeated in segment 5 and in E4. At five minutes rather than six, expect to take it.

---

2 · 13–17 min — Is mass action allowed in an *E. coli*?

Board 2 of 8 · 4 min · cumulative 9

Three columns on the board. Head them **well mixed**, **dilute**, **many**.

Well mixed — do this one live

$$\tau_{\mathrm{mix}} \sim \frac{L^2}{2D} = \frac{(1\ \mu\mathrm{m})^2}{2 \times 7.7\ \mu\mathrm{m}^2\,\mathrm{s}^{-1}} \approx 0.065\ \mathrm{s}$$

$$\tau_{\mathrm{rxn}} \sim \frac{1}{k_{\mathrm{on}} C} = \frac{1}{10^8\ \mathrm{M}^{-1}\mathrm{s}^{-1} \times 10^{-9}\ \mathrm{M}} \approx 10\ \mathrm{s}$$

$$\mathrm{Da} = \frac{\tau_{\mathrm{mix}}}{\tau_{\mathrm{rxn}}} \approx 0.007$$

> *Both numbers came from Tuesday. Say that.*
>
> ***Verdict: passes, by three orders of magnitude.*** *A molecule crosses the cell hundreds of times before it reacts, so there is no gradient to speak of.*
>
> ***IF ASKED*** *"is 10^8 right?" — that is diffusion-limited, the fastest possible. A typical protein–protein association is 10^5–10^6, which makes τ_{rxn}* ***longer*** *and Da* ***smaller****. The conclusion is robust in the direction that matters.*

Dilute — one line, no derivation

20% protein by weight. Excluded volume raises the activity of a large complex above its concentration; reported shifts run to about an order of magnitude for large complexes. **Verdict: bends.** k absorbs it — which is fine right up until you drop an *in vitro* k into a model of a cell without saying so.

(Mark this one as a range, not a measurement. It is the claim on this slide with the shortest shelf life.)

Many — the one that fails

$$1\ \text{nM} \approx 1\ \text{molecule per cell} \quad \Longrightarrow \quad N = 1, \qquad \frac{1}{\sqrt{N}} = 1$$

> ***Verdict: fails outright.*** *Not approximately — the quantity the ODE integrates does not describe any single cell.*

Land this: the three answers disagree, and not in the way anyone guesses. The spatial assumption — the one that *feels* shakiest in a crowded cell — is the one that passes most comfortably. What fails is counting, which is exactly what Session 2 said would fail.

> *Then the honest statement of the next ten weeks, in one sentence: we are going to integrate the mean of a process whose fluctuations are of order one, and it works because we mostly ask whether a design can work at all. Session 12 is where the mean stops being enough.*

---

3 · 17–21 min — The long way

Board 3 of 8 • 4 min • cumulative 13

Write the reaction, then the three derivatives — **and write them, do not show the slide**. The rhetorical move in segment 4 depends on the room having *felt* the repetition.

$$\mathrm{A} + \mathrm{B} \underset{k_r}{\overset{k_f}{\rightleftharpoons}} \mathrm{C}$$

$$\frac{d[\mathrm{A}]}{dt} = -k_f[\mathrm{A}][\mathrm{B}] + k_r[\mathrm{C}]$$
$$\frac{d[\mathrm{B}]}{dt} = -k_f[\mathrm{A}][\mathrm{B}] + k_r[\mathrm{C}]$$

> ***ASK*** *before writing the third line: what is $d[\mathrm{C}]/dt$? They will get it.* ***That is the point*** *— they already know the content, and what is coming is only bookkeeping.*

$$\frac{d[\mathrm{C}]}{dt} = +k_f[\mathrm{A}][\mathrm{B}] - k_r[\mathrm{C}]$$

Circle the two distinct fluxes. Say: **two fluxes, six terms, every term is one of those two with a sign.** Then: thirty species and forty reactions is where this course is by November — 1200 potential terms, each an opportunity for a sign error you will not find.

---

4 · 21–26 min — Two fluxes, one matrix

Board 4 of 8 · 5 min · cumulative 18

$$v_1 = k_f[\mathrm{A}][\mathrm{B}], \qquad v_2 = k_r[\mathrm{C}]$$

Draw the empty table. **Take the entries from the room, one row at a time.**

	v_1	v_2
A	-1	$+1$
B	-1	$+1$
C	$+1$	-1

> ***ASK*** *for the A row. Then the B row —* ***they will notice it is identical****, and that is worth naming out loud rather than passing over. Then C.*

Define it properly:

$$S_{ij} = \text{net molecules of species } i \text{ produced by reaction } j$$

$$\frac{d\mathbf{x}}{dt} = \mathbf{S}\,\mathbf{v}(\mathbf{x})$$

> ***Now multiply it out, row by row, out loud.*** *Row A gives $-v_1 + v_2$. Substitute the fluxes and the first equation from segment 3 reappears on the board next to it.* ***That moment is the session.*** *Do not rush it and do not narrate over it.*

Then the split, in two sentences: $\mathbf{S}$ is **structure** — a constant integer matrix off the reaction list, knowing nothing about rate constants, concentrations or time. $\mathbf{v}$ is **kinetics** — every parameter and all of the biology.

---

5 · 26–33 min — Both null spaces

Board 5 of 8 · 7 min · cumulative 25 · the longest, and the one to protect

Left first

$$\mathbf{w}^{\mathsf{T}}\mathbf{S} = \mathbf{0} \quad\Longrightarrow\quad \frac{d}{dt}\big(\mathbf{w}^{\mathsf{T}}\mathbf{x}\big) = \mathbf{w}^{\mathsf{T}}\mathbf{S}\mathbf{v} = 0 \quad \text{for any } \mathbf{v}$$

> *Three symbols, and it is the whole theorem. Say "for **any** v" twice — the result does not care about the kinetics at all.*
>
> ***ASK:** find me a $\mathbf{w}$. Give them thirty seconds. Expect $(1, 0, 1)$ and $(0, 1, 1)$.*

$$[\mathrm{A}] + [\mathrm{C}] \quad \text{and} \quad [\mathrm{B}] + [\mathrm{C}] \quad \text{are conserved}$$

Note the rank: the two columns of $\mathbf{S}$ are negatives of each other, so $\operatorname{rank}\mathbf{S} = 1$. Forward and reverse are not independent reactions. Hence

$$\dim(\text{left null}) = n - \operatorname{rank}\mathbf{S} = 3 - 1 = 2 \checkmark$$

Right next — this half is new to almost everyone

$$\mathbf{S}\mathbf{v} = \mathbf{0}, \quad \mathbf{v} \neq \mathbf{0}$$

> *Say the question in words first: **which patterns of reaction rates leave every concentration unchanged?***

For $\mathrm{A} + \mathrm{B} \rightleftharpoons \mathrm{C}$: $-v_1 + v_2 = 0$, so $v_1 = v_2$. One flux mode, and $\dim = r - \operatorname{rank}\mathbf{S} = 2 - 1 = 1$.

Then the cascade, as the contrast

$$\varnothing \xrightarrow{\alpha} m \qquad m \xrightarrow{\gamma_m} \varnothing \qquad m \xrightarrow{k_p} m + p \qquad p \xrightarrow{\gamma_p} \varnothing$$

$$\mathbf{S} = \begin{pmatrix} +1 & -1 & 0 & 0 \\ 0 & 0 & +1 & -1 \end{pmatrix}, \qquad \text{rank} = 2$$

$$\dim(\text{left null}) = 2 - 2 = 0, \qquad \dim(\text{right null}) = 4 - 2 = 2$$

> ***ASK before you say it:** no conservation laws at all — what does that mean physically? Answer: it is an **open system**. Matter is synthesised from nothing and degraded to nothing, so nothing can be conserved. That is not a defect in the model, it is the model being about a cell.*
>
> *The two flux modes: transcription balancing mRNA decay, and translation balancing protein decay. **The two independent balances, read straight off the matrix without solving anything**.*

Close the segment with one sentence and do not elaborate:

Flux balance analysis is the right null space, plus bounds on $\mathbf{v}$, plus an objective. That is all it is. Session 20.

IF ASKED *"how do you find these in practice?" — `scipy.linalg.null_space` on $\mathbf{S}$ and on $\mathbf{S}^\mathsf{T}$. One line each. Do not open a laptop here.*

CUT: the cascade contrast. Keep left, right, and the FBA sentence.

Then ConcepTest 1 at 33–37, off the board, and the pause at 37–39. The vote asks for the mRNA's net stoichiometry in m → m + p; the pause asks them to write S for the cascade, where that is the third column. Do not let the ConcepTest overrun — the adjacency is the point.

---

6 · 59–63 min — The steady state, without a computer

Board 6 of 8 · 4 min · cumulative 29

$$\frac{dm}{dt} = \alpha - \gamma_m m = 0 \quad\Longrightarrow\quad m^* = \frac{\alpha}{\gamma_m}$$

The move worth naming: ***a cascade solves top-down.*** *m^* does not contain p, so solve it alone and substitute. First appearance of a structure they will use constantly.*

$$\frac{dp}{dt} = k_p m^* - \gamma_p p = 0 \quad\Longrightarrow\quad p^* = \frac{k_p \alpha}{\gamma_m \gamma_p}$$

Numbers: $\alpha = 10$, $\gamma_m = 0.5$, $k_p = 4$, $\gamma_p = 0.05$ give $m^* = 20$ and $p^* = 1600$.

IF ASKED / IF OFFERED *"p^* doesn't depend on the initial conditions at all" — correct, and it is Session 8's territory: one globally stable fixed point. Say the phrase, move on.*

---

7 · 63–67 min — Integrate it exactly

Board 7 of 8 · 4 min · cumulative 33

The only closed-form integration in the course. Do every line.

$$m(t) = m^* \left(1 - e^{-\gamma_m t}\right)$$

$$\frac{dp}{dt} + \gamma_p p = k_p m(t)$$

Multiply by the integrating factor $e^{\gamma_p t}$:

$$\frac{d}{dt}\left[p\, e^{\gamma_p t}\right] = k_p m(t)\, e^{\gamma_p t}$$

> *Say:* ***this is the move****, and it comes back in Session 5 for response time and Session 11 for the delayed oscillator.*

Integrate and rearrange:

$$p(t) = p^* \left[1 - \frac{\gamma_m e^{-\gamma_p t} - \gamma_p e^{-\gamma_m t}}{\gamma_m - \gamma_p}\right]$$

> ***CHECK 1, out loud:*** *at $t = 0$ the bracket is $(\gamma_m - \gamma_p)/(\gamma_m - \gamma_p) = 1$, so $p(0) = 0$. ✓*
>
> ***CHECK 2:*** *as $t \to \infty$ both exponentials die and $p \to p^*$. ✓*
>
> *Ten seconds each, and they are the habit being taught.*

Two exponentials, not one. The protein does not simply relax at γ_p; it also carries the mRNA's rise.

> ***IF ASKED*** *"what if $\gamma_m = \gamma_p$?" — somebody always asks. It is a* ***removable singularity****; the limit exists, $p = p^* \left[1 - (1 + \gamma t)e^{-\gamma t}\right]$, and physically it is the critically damped case: the two relaxation modes have collided. Comes back as repeated eigenvalues in Session 8.* ***If nobody asks, say it anyway.***

---

8 · 70–74 min — Four parameters become one

Board 8 of 8 • 4 min • cumulative 37

$$\tau = \gamma_p t, \qquad \mu = \frac{m}{m^*}, \qquad \pi = \frac{p}{p^*}$$

Derive both, two lines each — do not display them.

$$\frac{dm}{dt} = \alpha - \gamma_m m \Big/ m^* \implies \frac{d\mu}{dt} = \gamma_m(1 - \mu) \xrightarrow{\tau = \gamma_p t} \frac{d\mu}{d\tau} = \frac{1}{\epsilon}(1 - \mu)$$

$$\frac{d\pi}{d\tau} = \mu - \pi, \qquad \boxed{\epsilon = \frac{\gamma_p}{\gamma_m}}$$

Land the parameter count, plainly: you started with four parameters and you now have one. Two cascades with wildly different rate constants and the same ϵ give the *same curve* after rescaling — so **no experiment measuring the shape alone can tell them apart.** That is a real and slightly unwelcome result; let the 247 half sit with it.

Then the punchline, and hand straight to the slide:

$$\epsilon \to 0 \implies \mu = 1 \text{ instantly } = \textbf{the quasi-steady-state approximation}$$

Ours is $\gamma_p/\gamma_m = 0.05/0.5 = 0.1$.

> ***IF ASKED*** *whether this has a name — Buckingham Pi, without the ceremony. Say so only if someone raises it; do not introduce it.*

If you are running late

Take them in this order. Each is chosen so nothing downstream breaks.

1. **The dimer paragraph in segment 1** (30 s). Repeated in segment 5 and E4.
2. **The cascade contrast in segment 5** (2 min). Keep left null, right null and the FBA sentence — those three are what Session 20 needs.
3. **CHECK 2 in segment 7** (20 s). Keep CHECK 1; one sanity check taught properly beats two done at speed.
4. **The degeneracy discussion in segment 7** (30 s) — *only* if nobody asked.

Do **not** cut segment 8. It is what makes Session 4 a derivation rather than a recipe, and it is four minutes.